INCH-POUND
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DETAIL SPECIFICATION SHEET

# SWITCH, TOGGLE, POSITIVE BREAK, MINIATURE, TOGGLE SEALED, PRINTED CIRCUIT BOARD TERMINALS, SINGLE POLE, .250 MOUNTING BUSHING

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the products described herein shall consist of this specification sheet and MIL-DTL-8834.

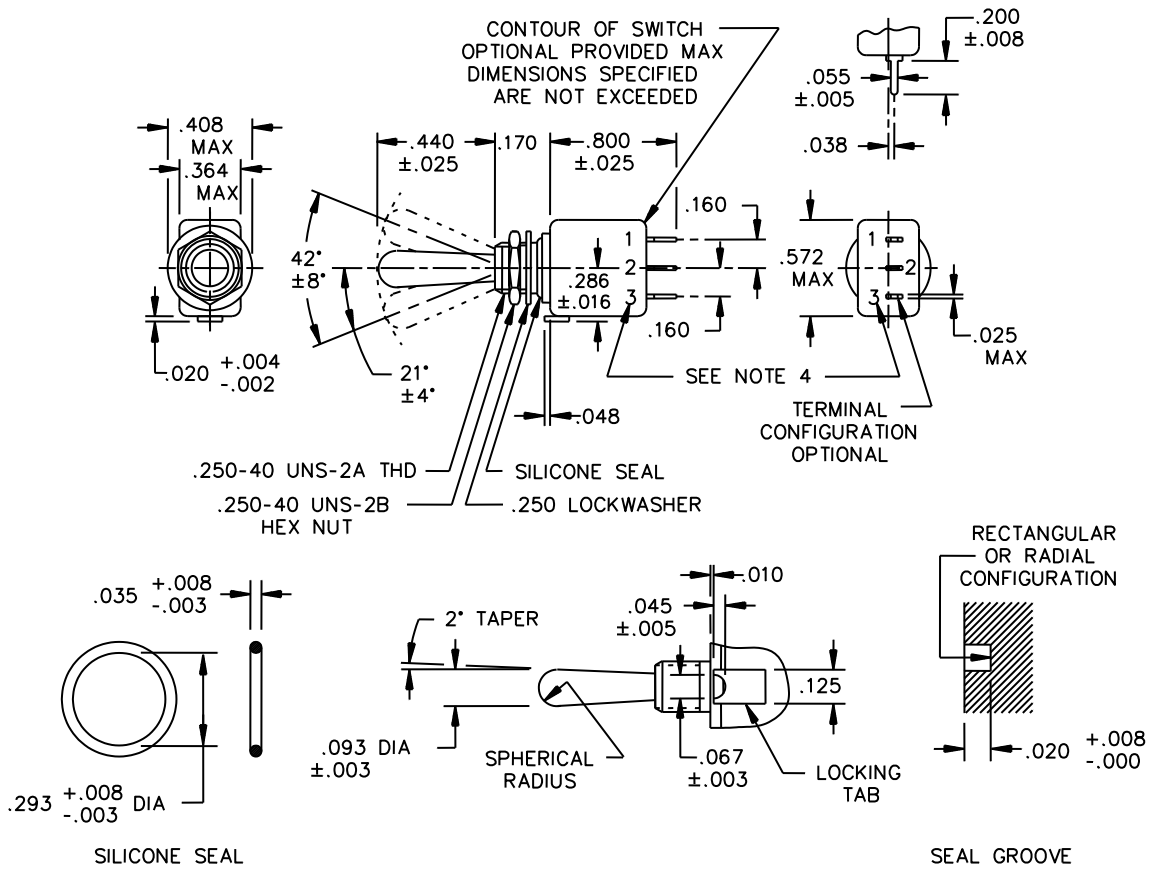
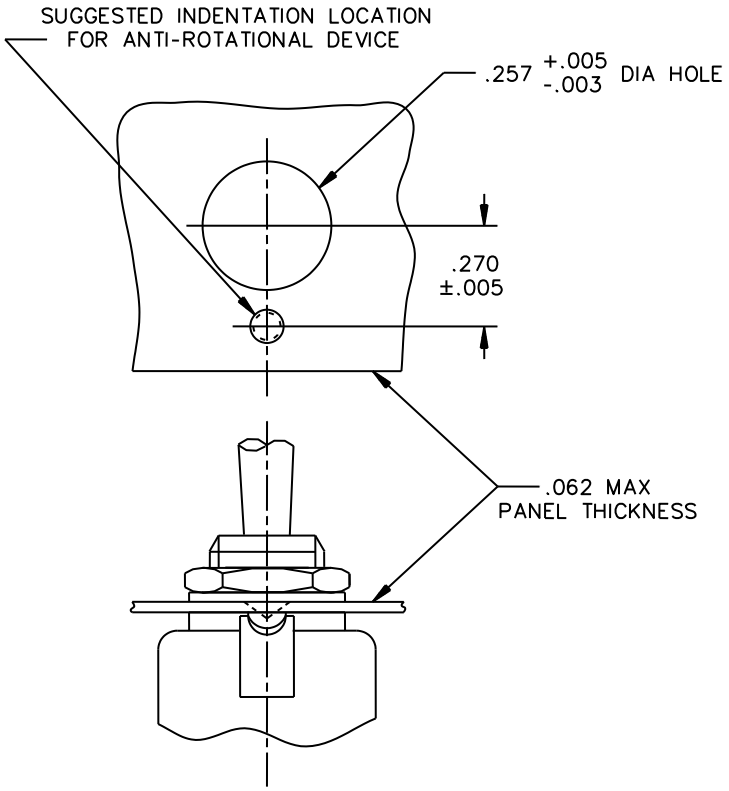



FIGURE 1. Dimensions and configuration.



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| Inches | mm | Inches | mm | Inches | mm | Inches | mm |
|---|---|---|---|---|---|---|---|
| .002 | 0.05 | .025 | 0.64 | .093 | 2.36 | .286 | 7.26 |
| .003 | 0.08 | .035 | 0.89 | .125 | 3.18 | .293 | 7.44 |
| .004 | 0.10 | .038 | 0.97 | .160 | 4.06 | .364 | 9.25 |
| .005 | 0.13 | .045 | 1.14 | .170 | 4.32 | .408 | 10.36 |
| .008 | 0.20 | .048 | 1.22 | .200 | 5.08 | .440 | 11.18 |
| .010 | 0.25 | .055 | 1.40 | .250 | 6.35 | .572 | 14.53 |
| .016 | 0.41 | .062 | 1.57 | .257 | 6.53 | .800 | 20.32 |
| .020 | 0.51 | .067 | 1.70 | .270 | 6.86 | | |

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerances are ±.010 (0.25 mm) on decimals and ±5° on angles.
4. Terminals need not be marked. Terminal identification is shown for reference purposes only.

FIGURE 1. Dimensions and configuration - Continued.



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REQUIREMENTS:

For hardware detail specifications, see appendix of MIL-DTL-8834.

In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

Dielectric withstanding voltage: For center on circuits, DWV shall be 1,200 V rms at sea level.

115 V ac, 60 hertz electrical endurance tests are to performed at room temperature and pressure.

Switches shall be rigidly mounted by their normal means on a rigid metal panel and subjected to shock test, of MIL-STD-202-207. The switch shall be electrically and mechanically operative at the conclusion of the test and there shall be no mechanical transfer during the test.

Terminals: Gold plated.

Suitable for mounting on panels .062 inches (1.57 mm) maximum thickness.

Weight: .011 pound maximum (5 grams).

Strength of terminals: 5 pounds normal to the mounting plane and 1 pound in other planes.

Altitude: 50,000 feet.

Strength of actuator: Lever pivot and lever stop 6 pounds.

TABLE I. Detail requirements.

| MS dash no | Circuit made between terminals as indicated with the toggle lever in these positions: 1/ | | | Current capacity in amperes | | | | | | Low current level switching 5 mV |
|---|---|---|---|---|---|---|---|---|---|---|
| | | | | Resistive load | | | Inductive load | | | |
| | | | | 28 volts dc | 115 volts | | 28 volts dc | 115 volts | | |
| | Opposite locking tab side 4/ | Center position | Locking tab side 4/ | | 60 Hertz ac | 400 Hertz ac | | 60 Hertz ac | 400 Hertz ac | |
| -211 | on 2-3 | off | on 1-2 | 5 | 2 | 3 | 1 2/ | 1 | 2 | 25 µA 3/ |
| -221 | | none | off | | | | | | | |
| -231 | | | on 1-2 | | | | | | | |
| -241 | | off | none | | | | | | | |
| -271 | mom-on 2-3 | | mom-on 1-2 | | | | | | | |
| -281 | none | | | | | | | | | |
| -311 | on 2-3 | | | | | | | | | |
| -321 5/ | none | on 2-3 | | | | | | | | |

1/ Direction of movement of internal mechanism is opposite to the direction of the toggle movement.
2/ With time constant of .020 ±.002 second.
3/ Contact resistance not to exceed 50Ω during life, low current level switching.
4/ Non-functional terminals shall not be supplied.
5/ Delayed action of the switch toggle lever may cause circuit to close or open before snap action mechanism trips.



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Referenced documents:
MIL-DTL-8834
MIL-STD-202-207

Changes from previous issue. The margins of this specification are marked with vertical lines to indicate modifications generated by this amendment. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations.

Custodians:
Army - CR
Navy - AS
Air Force - 85
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Review activities:
Army - AR
Navy - EC
Air Force - 99

Preparing activity:
DLA – CC

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